

# Customer Retention & Churn Risk Analysis

**Prepared for:** Executive Leadership, Marketing & Customer Success

**Data Source:** Online Retail transaction data (UK-based retailer, 1 Dec 2010 – 9 Dec 2011)

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## 1. Executive Summary

Twelve months of transaction data (£10.64M revenue, 19,962 transactions, 4,338 customers) were cleaned, segmented, and modeled to identify which customers are about to stop buying — before they actually do.

Two facts anchor the entire analysis:

- **Retention, not acquisition, drives the business.** 65.6% of customers are repeat buyers, and they generate 93.1% of customer revenue — at roughly 7x the value of a one-time buyer.
- **A simple, three-feature Logistic Regression model reliably separates customers by real churn risk.** Its top-risk tier churns at **61.5%**; its bottom-risk tier churns at just **8.4%** — a 7.3x spread, with a ROC-AUC of **0.74**.

Applied to the test population, the model's **340 High Risk customers (51% of those scored)** hold roughly **13% of at-risk revenue exposure**, and independently cross-check against RFM segmentation: 99% of "Hibernating" customers and 40–86% of "At Risk," "Other," and "Recent" RFM segments are confirmed High Risk by the model. This agreement between two independent methods is the strongest signal in the report — it tells leadership exactly where to spend, and where *not* to.

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## 2. Business Problem & Objective

The retailer has no systematic way to flag customers who are about to churn *before* they go quiet. With **34.4% of the customer base never returning after one purchase**, and repeat customers carrying almost all customer revenue, undetected attrition among mid-and-high-value customers is a direct, preventable revenue loss.

**Objective:** Build a forward-looking churn definition (does a customer return within the next 90 days?), segment customers by behavior (RFM), and train a model that ranks customers by real retention risk — accurate enough to prioritize retention spend rather than applying it evenly.

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### 3. Methodology Snapshot

| Stage            | What Was Done                                                                                                                                                                                                              |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data cleaning    | 541,909 raw transaction lines → 524,878 clean sales records after removing duplicates (0.97%), cancelled orders, and invalid prices/quantities.<br>Customer-level analysis uses 392,692 rows / 4,338 identified customers. |
| Segmentation     | RFM quintile scoring (Recency, Frequency, Monetary) → 6 behavioral segments.                                                                                                                                               |
| Churn definition | <b>Retained = 1</b> if a customer purchased again within Sep–Nov 2011 (90-day forward window); features built only from data through 31 Aug 2011 to avoid leakage.                                                         |
| Model            | Logistic Regression, 3 features (Historical Orders, Historical Recency, Customer Tenure), 80/20 stratified train/test split (2,653 / 664 customers), features standardized before fitting.                                 |
| Evaluation       | Standard classification metrics (Accuracy, Precision, Recall, F1, ROC-AUC) to evaluate a Logistic Regression classifier and translate its coefficients into business-interpretable odds ratios.                            |
| Validation       | Metrics on held-out test set; model risk tiers cross-checked against independent RFM segments (strong agreement).                                                                                                          |

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## 4. The Role of SQL

SQL (via DuckDB, run inside the notebooks) served as the project's audit layer — independently re-deriving every key metric in parallel with Pandas, which is why figures like £10.64M total revenue and 4,338 unique customers can be reported with confidence rather than caveats.

That cross-check also caught a real bug: "InvoiceNo" stored as mixed integer/string types was silently distorting per-customer order counts — the single strongest driver in the churn model (odds ratio 3.51). Fixing it prevented the model from training on corrupted data.

SQL was also used to cleanly separate "known before 31 Aug 2011" from "what happened Sep–Nov 2011" when building the churn features, guarding against data leakage that would have inflated the model's reported accuracy.

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## 5. Key Business Trends

**1. Repeat customers are the business.** 2,845 repeat customers (65.6% of the base) generate £8.27M (93.1%) of customer revenue at an average £2,908/customer — versus £411/customer for one-time buyers. Repeat customers also buy across ~4x more distinct products (82 vs. 22 on average), showing that product-range engagement and loyalty move together.

**2. Revenue is concentrated in customers, not products.** The top 10% of customers (434 people) generate 61.5% of total revenue — a classic Pareto pattern. Product revenue, by contrast, is *not* concentrated (top 10 products = only 10.8% of revenue), so customer retention — not merchandising a handful of hero SKUs — is the higher-leverage lever.

**3. "At Risk" is a large, recoverable pool, not a lost cause.** 903 customers (RFM) previously spent an average £1,274 and ordered nearly 3 times, but haven't purchased in ~159 days. The model confirms this ambiguity: roughly 40% of them score High Risk and 51% Medium Risk — an active win-back window, not a dead segment.

**4. Order frequency is the single strongest churn driver — well ahead of recency or tenure.** In the model, a one-standard-deviation increase in prior order count more than triples the odds of retention (odds ratio **3.51**), roughly 2.5x the effect size of tenure (1.27) and far outweighing the (expected, negative) effect of recency (0.75). The clearest business lever is converting a first-time buyer into a second-time buyer quickly — not simply keeping a customer "on the books" longer.

**5. RFM alone misses early defection inside your best customers.** 61.5% of "Champions" score Low Risk as expected — but **15.5% already score High Risk and 23% Medium Risk**,

despite looking safe on a traditional RFM view. The model catches this; RFM segmentation by itself does not.

**6. Hibernating customers are a confirmed dead end for retention spend.** 98.8% of the 824-customer "Hibernating/Low Engagement" RFM segment is independently flagged High Risk by the model (avg. 227 days since last purchase) — near-total agreement between two unrelated methods. Marginal retention budget here has low expected payoff.

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## 6. Model Performance & What Drives Churn

Performance (held-out test set, n=664):

| Metric               | Value |
|----------------------|-------|
| Accuracy             | 68.2% |
| Precision (Retained) | 75.3% |
| Recall (Retained)    | 65.1% |
| ROC-AUC              | 0.739 |

The model's real value shows up in its **risk tiers**, not the raw accuracy number:

| Risk Tier                               | Customers | Actual Churn Rate |
|-----------------------------------------|-----------|-------------------|
| <b>High Risk</b> (<50% retention prob.) | 340       | <b>61.5%</b>      |
| Medium Risk (50–70%)                    | 169       | 39.6%             |
| <b>Low Risk</b> (≥70%)                  | 155       | <b>8.4%</b>       |

A 7.3x gap between the top and bottom tiers means retention spend targeted at High Risk customers is dramatically more efficient than spreading it evenly across the base.

**What drives the score, in plain terms:**

| Driver                          | Odds Ratio | Meaning                                                                    |
|---------------------------------|------------|----------------------------------------------------------------------------|
| Historical order count          | 3.51       | The single biggest lever — more historical orders strongly predicts return |
| Customer tenure                 | 1.27       | Longer relationships help retention, but modestly                          |
| Recency (days since last order) | 0.75       | Every step further from last purchase lowers retention odds                |

## 7. Financial Impact & Strategic Recommendations

**Impact:** In the scored test cohort, High Risk accounts hold roughly £220K (13% of that cohort's revenue) at a 61.5% churn rate, versus 8.4% for Low Risk — the gap that makes targeted retention spend pay off. The At-Risk RFM segment alone represents an estimated **£1.15M in recoverable historical customer value** (903 customers × £1,274 average spend), the single largest realistic win-back opportunity in the dataset.

### Recommendations :

1. **Make the model's risk score an active trigger, not a report.** Customers crossing into High Risk should automatically enter a win-back workflow (targeted offer, personal outreach) — the model catches 72% of true churners in time to act.
2. **Prioritize the At-Risk RFM segment first.** Highest combination of recoverability and historical value; unlike Hibernating customers, they're recent enough to still respond.
3. **Build a "second purchase" acceleration program.** Since order frequency is the dominant churn driver, the highest-leverage investment is converting first-time buyers into repeat buyers fast (time-limited second-purchase incentive, early loyalty rewards).
4. **Use model scores, not RFM alone, to protect Champions.** ~38% of Champions already show elevated risk that RFM segmentation misses — route VIP retention outreach off the model, not the segment label.
5. **De-prioritize Hibernating/Low Engagement spend.** Near-unanimous High Risk confirmation with no recent purchase signal means this budget is better redeployed to At Risk and Champions.